
TECHNICAL ASSESSMENT PRESENTATION

CareLink National EHR

Architecture, Delivery
& Resilience

EHR

NATIONAL HEALTH RECORD

Continuity
Integrity
Scale

CareLink must work where connectivity does not

1,500

FACILITIES

National footprint

~3,000

CONCURRENT USERS

Weekday-morning peak

1,100

CONNECTIVITY CONSTRAINED

800 intermittent + 300 offline periods

DESIGN PRIORITIES

01 Clinical continuity

02 Data integrity

03 Security & privacy

04 Operational
simplicity

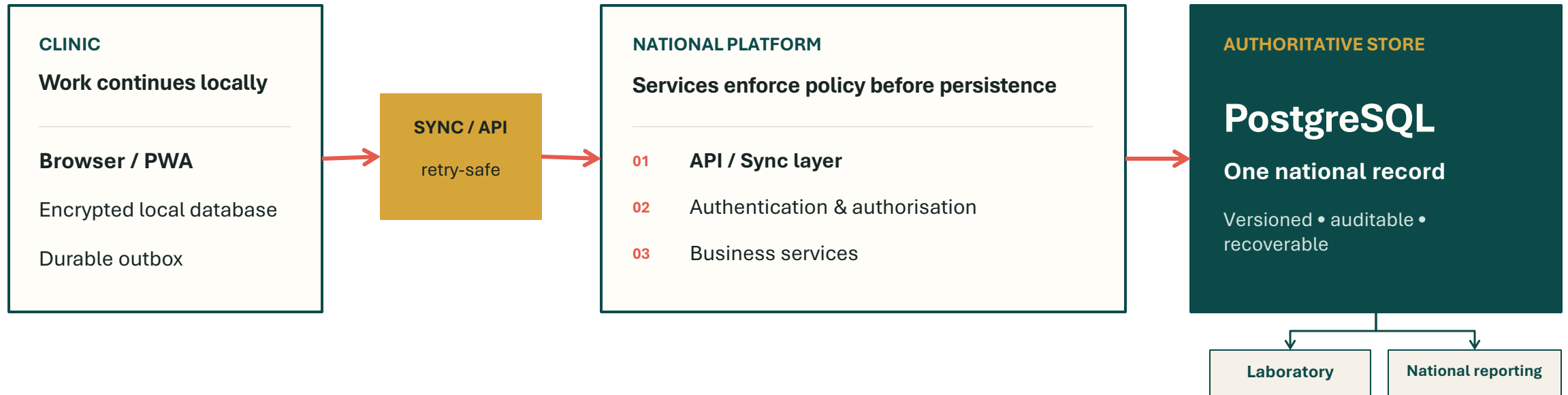
OPERATING CONSTRAINTS

Older clinic hardware and browsers
Sensitive patient information

ASSUMPTION

A visit before the missed appointment does not clear
follow-up; a visit after it does.

Offline work synchronises safely into one national source of truth



CROSS-CUTTING CONTROLS

IDENTITY

OIDC / OAuth2

Role + facility / district scope

Time-limited offline sessions

INTEGRITY

Durable outbox + operation IDs

Cursor-based synchronisation

Explicit version conflicts

OPERATIONS

Sync health + queue age

Failed synchronisations

Availability + reporting freshness

Three choices reduce clinical risk without importing needless complexity

		CHOSEN	REJECTED	WHY IT HOLDS
01	Offline-first CONTINUITY	Local work + durable synchronisation	Online-only	Connectivity cannot become a clinical dependency when 1,100 facilities cannot rely on continuous connectivity.
02	Modular monolith first SIMPLICITY	Well-separated modules in one deployable platform	Immediate microservices	The scale does not justify distributed-system complexity on day one.
03	Idempotent mutations INTEGRITY	Every mutation carries an operation ID	Every retry treated as a new request	A timeout does not reveal whether the write committed; a retry must be safe.

Local retention should be governed by clinical need—not device capacity

CURRENT POSITION

Keep only the minimum necessary facility data

- Encrypted local database
- OS/device-protected encryption keys
- Configurable stale-data retention
- User revalidation when connectivity returns
- No national patient population cached locally

WHAT WOULD CHANGE THE DECISION?

Evidence—not assumption—sets the retention boundary

- 01 Actual maximum offline duration
- 02 Clinical workflows
- 03 Ministry privacy/security requirements
- 04 Device-management capability
- 05 Data sensitivity and volume

PRINCIPLE Do not keep sensitive data locally simply because storage is available.

The vertical slice proves the workflow — not production readiness

BACKEND	FRONTEND	API + EVIDENCE	STATUS
• Django + Django REST Framework • PostgreSQL • Typed API responses • Role/facility authorisation • Server-side pagination • Filtering and sorting • Correlation IDs + structured errors • Transactional contact recording	• React + TypeScript • API service layer • Facility/status filters • Days-overdue sorting • Pagination • Loading / empty / error / retry • Accessible semantic controls • Contact workflow	• GET /api/follow-up • POST /api/follow-up/{id}/contacted • 13 backend tests passing • Frontend lint ✓ • Production build ✓	WORKING Core follow-up vertical slice

DESIGNED NEXT

01 AUTH	02 OFFLINE	03 WRITES	04 SYNC	05 INTEGRATE	06 OPS
OIDC / OAuth2	Encrypted store + outbox	Idempotency + retry	Versioned conflicts	Lab + reporting boundaries	In-country deploy + monitoring

Security and clinical integrity block merge before optimisation does

THREE CATEGORIES OF RISK

CRITICAL SECURITY	• Hard-coded database credentials • Missing authorisation	• SQL injection • PHI / National ID in logs
CRITICAL CLINICAL DATA INTEGRITY	• Swallowed exceptions • Unsafe SQL writes	• Non-transactional writes • Incorrect follow-up evaluation
MAJOR SCALABILITY & UX	• N+1 database queries • 3-second frontend polling • Direct state mutation	• No pagination • Client-side filtering • Non-semantic interactive controls

THREE MERGE BLOCKERS

01 Security & authorisation

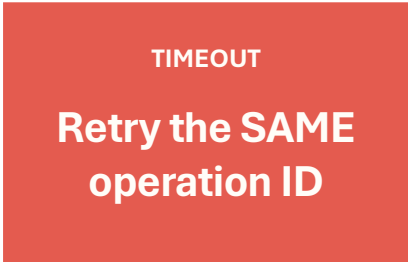
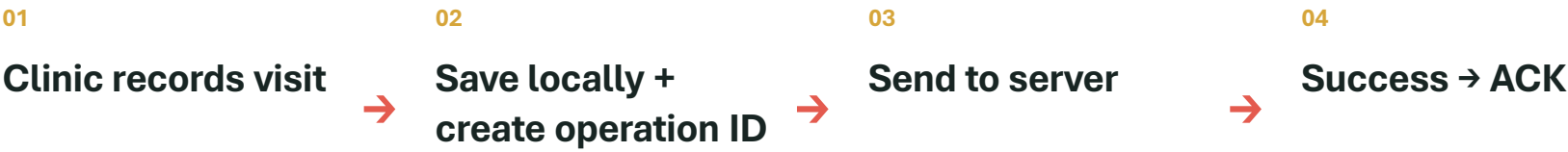
02 Clinical data integrity

03 Scalability & follow-up correctness

Merge only after the evidence shows these risks are controlled.

A timeout is not the same as a failure

THE KEY RULE



CONFLICT HANDLING



DUPLICATE RECORDS

Detect → Preserve → Review → Merge / Canonicalise

PRINCIPLE Never casually delete clinical records.

Next week turns a working slice into production evidence

01 DAYS 1–2

- Harden authentication + authorisation
- Frontend/API automated tests
- Accessibility review
- API contract validation

02 DAYS 3–5

- Durable client outbox
- Idempotency store
- Retry / backoff
- Cursor-based synchronisation
- Conflict UX

03 DAYS 6–7

- 10k+ follow-up performance test
- Concurrent-user/load validation
- Offline/replay testing
- Security checklist
- Deployment + rollback rehearsal

GOAL

Not “more code” — evidence that the system is safe to scale.

Questions

Architecture · Delivery · Leadership

Continuity
Integrity
Scale